

GED Science Practice Test 16

PART A — QUESTIONS

GED-SCI16-001

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A researcher studies whether a guided breathing routine affects resting heart rate. Group A completes 10 minutes of slow guided breathing each day for two weeks. Group B sits quietly for the same amount of time without breathing instructions. Resting heart rate is measured at the end of the study.

Group | Average resting heart rate (beats/min)

A | 68

B | 75

Which conclusion is best supported by the data?

A) Guided breathing was associated with a lower average resting heart rate.

B) Guided breathing was associated with a higher average resting heart rate.

C) Both groups had exactly the same resting heart rate.

D) Resting heart rate cannot be measured in beats per minute.

GED-SCI16-002

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares the momentum of four moving objects. Momentum can be calculated as $\text{mass} \times \text{velocity}$.

Object | Mass (kg) | Velocity (m/s)

A | 3 | 6

B | 5 | 4

C | 8 | 2

D | 2 | 9

Which object has the greatest momentum?

A) Object A

B) Object B

C) Object C

D) Object D

GED-SCI16-003

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A scientist compares two farming methods. In one method, fields are left bare after harvest. In another method, cover crops are planted between growing seasons. Cover crops can add organic matter, reduce soil compaction, and help soil hold nutrients.

Which outcome would best support the use of cover crops?

- A) Fields with cover crops show higher soil organic matter after several seasons.
- B) Fields with cover crops stop receiving sunlight.
- C) Cover crops turn soil into igneous rock.
- D) Cover crops prevent all seasonal weather changes.

GED-SCI16-004

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A person's pupils change size in response to light. In bright light, muscles in the iris make the pupil smaller, reducing the amount of light entering the eye. In dim light, the pupil becomes larger so more light can enter.

Which body system coordinates this rapid response?

- A) Nervous system
- B) Digestive system
- C) Skeletal system
- D) Excretory system only

GED-SCI16-005

Section: Science

Type: numeric_entry

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A 6 kg object moves at 7 m/s.

Momentum = mass $\times$ velocity

What is the object's momentum in kg·m/s?

Type your answer as a whole number.

GED-SCI16-006

Section: Science

Type: multi_select

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A student tests how the mass of a cart affects its momentum when all carts move at the same speed. The student uses carts of different masses and releases each cart so it reaches 2 m/s.

Which TWO factors should be kept constant?

- A) Cart speed
- B) Track surface
- C) Cart mass
- D) Calculated momentum
- E) Final result

GED-SCI16-007

Section: Science

Type: dropdown

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A person with one changed copy of a gene may not show symptoms, but if that person has children with another carrier, a child may inherit two changed copies. This pattern is consistent with [D1], and the person with one changed copy is called a [D2].

Dropdowns:

D1) recessive inheritance | contact metamorphism | wave reflection

D2) carrier | fulcrum | aquifer

GED-SCI16-008

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student hears a siren from an ambulance as it moves past. The pitch sounds higher while the ambulance approaches and lower after it passes. The sound waves are compressed in front of the moving ambulance and stretched behind it.

Which wave phenomenon is described?

- A) Doppler effect
- B) Chemical neutralization
- C) Sublimation
- D) Chromosome separation

GED-SCI16-009

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

Ocean scientists study magnetic patterns in rocks on both sides of a mid-ocean ridge. The patterns form matching stripes. New crust forms at the ridge and moves away over time, recording Earth's magnetic field as it cools.

Which conclusion is best supported by this evidence?

- A) Seafloor spreading occurs at mid-ocean ridges.
- B) Ocean crust forms only from compacted shells.
- C) Earth's magnetic field prevents all rock formation.
- D) Mid-ocean ridges are caused by river flooding.

GED-SCI16-010

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A medical lab measures blood oxygen level before and after a breathing treatment.

Condition | Average blood oxygen saturation (%)

Before treatment | 91

After treatment | 96

Which statement is best supported by the data?

- A) Average blood oxygen saturation increased after treatment.
- B) Average blood oxygen saturation decreased after treatment.
- C) Oxygen saturation stayed exactly the same.
- D) The treatment made oxygen disappear from the blood.

GED-SCI16-011

Section: Science

Type: drag_drop

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Match each target with the correct tile.

Tiles:

- A) Type of wheel tread used on the model vehicle
- B) Distance the vehicle travels before stopping
- C) Same starting speed in each trial
- D) Color of the vehicle body

Targets:

- T1) Independent variable
- T2) Dependent variable
- T3) Controlled variable

GED-SCI16-012

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

The gallbladder stores bile made by the liver. After a fatty meal, bile is released into the small intestine. Bile helps break large fat droplets into smaller droplets, making it easier for digestive enzymes to act.

Which statement best describes the role of bile?

- A) It helps physically break fat into smaller droplets for digestion.
- B) It carries nerve signals across synapses.
- C) It stores chromosomes during meiosis.
- D) It changes oxygen into carbon dioxide in the lungs.

GED-SCI16-013

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student rubs two different fabrics together and notices that one fabric clings to the other. Electrons moved from one material to the other, leaving the materials with opposite electric charges.

Which interaction best explains the clinging?

- A) Opposite electric charges attract.
- B) The fabrics became radioactive elements.
- C) The fabrics stopped having mass.
- D) Gravity disappeared between the fabrics.

GED-SCI16-014

Section: Science

Type: numeric_entry

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A reforestation project increases tree-covered land from 1,200 hectares to 1,500 hectares.

What is the percent increase in tree-covered land?

Type your answer as a percent rounded to the nearest whole number.

GED-SCI16-015

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A researcher studies whether resistance training affects bone strength. One group completes supervised resistance training for 12 weeks. A comparison group continues normal daily activity. Both groups have the same bone-strength measurement before and after the study.

Why should both groups be measured before and after the study?

- A) To compare changes in bone strength over time.
- B) To guarantee that both groups become identical.
- C) To prevent the comparison group from being useful.
- D) To make bones stop responding to exercise.

GED-SCI16-016

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student tests how the surface material of a slide affects the speed of a block at the bottom. The student uses the same block and same slide height for every trial. The surface material is changed.

What is the independent variable?

- A) Surface material of the slide
- B) Speed of the block at the bottom
- C) Same block
- D) Same slide height

GED-SCI16-017

Section: Science

Type: dropdown

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A ramp allows a worker to lift a heavy object using less input force over a longer distance. The ramp is an [D1], and it can provide [D2].

Dropdowns:

D1) inclined plane | ribosome | glacier

D2) mechanical advantage | genetic variation | protein synthesis

GED-SCI16-018

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A doctor tracks average blood pressure after patients begin a supervised walking program.

Time | Average systolic blood pressure (mm Hg)

Start | 146

4 weeks | 138

8 weeks | 132

12 weeks | 128

Which statement is best supported by the data?

- A) Average systolic blood pressure decreased over the 12 weeks.
- B) Average systolic blood pressure increased over the 12 weeks.
- C) Blood pressure stayed exactly the same.
- D) Walking made blood pressure drop to zero.

GED-SCI16-019

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares the distance a sled travels on four surfaces after the same push.

Surface | Distance traveled (m)

Packed snow | 9.6

Short grass | 4.2
Dry pavement | 2.8
Rubber mat | 1.3

Which surface most likely produced the least friction?

- A) Packed snow
- B) Short grass
- C) Dry pavement
- D) Rubber mat

GED-SCI16-020

Section: Science

Type: multi_select

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

Select TWO answers.

A region tracks average glacier thickness and average summer temperature.

Year | Average glacier thickness (m) | Average summer temperature (°C)

1 | 92 | 10.8

2 | 86 | 11.2

3 | 79 | 11.7

4 | 71 | 12.3

Which TWO statements are best supported by the data?

- A) Average glacier thickness decreased from Year 1 to Year 4.
- B) Average summer temperature increased from Year 1 to Year 4.
- C) Glacier thickness increased every year.
- D) Summer temperature was lowest in Year 4.
- E) Both measurements stayed exactly the same.

GED-SCI16-021

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Some hormones are released in pulses rather than at a constant rate. For example, growth hormone may be released in larger pulses during deep sleep. These pulses can signal body tissues to support growth and repair.

Which statement best describes one role of growth hormone?

- A) It can signal tissues involved in growth and repair.
- B) It forms minerals in igneous rock.
- C) It bends sound waves around barriers.
- D) It carries oxygen as a red blood cell.

GED-SCI16-022

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student pushes a heavy cabinet across the floor. The cabinet moves at a constant speed because the applied push is equal in size and opposite in direction to friction.

Which statement best describes the forces on the cabinet?

- A) The forces are balanced, so the cabinet moves at constant speed.
- B) The forces are unbalanced, so the cabinet must accelerate.
- C) Friction acts in the same direction as the push.
- D) The push removes the cabinet's weight.

GED-SCI16-023

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A small rocky object enters Earth's atmosphere. Friction with air heats its surface, producing a bright streak of light. If part of the object reaches the ground, the remaining piece is called a meteorite.

What is the object called while it is producing a bright streak in the atmosphere?

- A) Meteor
- B) Aquifer
- C) Folded mountain
- D) Humus layer

GED-SCI16-024

Section: Science

Type: numeric_entry

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A clinic reviews 450 records. Of these records, 72 show a documented thyroid condition.

What percent of the records show a documented thyroid condition?

Type your answer as a percent rounded to the nearest whole number.

GED-SCI16-025

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student compares how different helmet liners affect peak force during a model impact. The same model head is dropped from the same height with each liner. A sensor records peak force during impact.

Which factor is the dependent variable?

- A) Peak force recorded by the sensor
- B) Type of helmet liner
- C) Same model head
- D) Same drop height

GED-SCI16-026

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Mitochondria release usable energy from food molecules during cellular respiration. Muscle cells often contain many mitochondria because muscle contraction requires large amounts of usable energy.

Which statement best describes mitochondria?

- A) They help cells release usable energy from food molecules.
- B) They store groundwater in underground rock layers.
- C) They make chromosomes separate during earthquakes.
- D) They reflect sunlight from the Moon.

GED-SCI16-027

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student tests how a lever's input distance relates to the output distance of a load.

Trial | Input distance moved (cm) | Output distance moved (cm)

- A | 10 | 5
- B | 20 | 10
- C | 30 | 15
- D | 40 | 20

Which statement is best supported by the data?

- A) The output distance was half the input distance in each trial.
- B) The output distance was twice the input distance in each trial.
- C) The output distance stayed the same in every trial.
- D) Input distance had no relationship to output distance.

GED-SCI16-028

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares the average distance of four objects from the Sun.

Object | Average distance from Sun (million km)

A | 108

B | 150

C | 228

D | 778

Which object is farthest from the Sun?

A) Object A

B) Object B

C) Object C

D) Object D

GED-SCI16-029

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A lab measures average muscle recovery time after the same exercise.

Group | Average recovery time (hours)

Stretching routine | 18

No stretching routine | 25

Which claim is best supported by the data?

A) The stretching group had a shorter average recovery time.

B) The no-stretching group had a shorter average recovery time.

C) Both groups had the same average recovery time.

D) Stretching made recovery time impossible to measure.

GED-SCI16-030

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student uses polarized sunglasses to reduce glare from reflected sunlight. The lenses allow light vibrating in some directions to pass while blocking much of the horizontally reflected light.

Which wave property is being used?

A) Polarization

B) Boiling point

C) Blood pressure

D) Plate tectonics

GED-SCI16-031

Section: Science

Type: multi_select

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A researcher studies whether music tempo affects walking pace. Volunteers walk on the same indoor track while listening to slow, medium, or fast tempo music. Walking pace is measured for each condition.

Which TWO steps would improve the investigation?

- A) Use the same track for all walking trials.
- B) Measure walking pace the same way each time.
- C) Change the track surface for each music tempo.
- D) Record only the fastest walking pace.
- E) Let each volunteer use a different timing method.

GED-SCI16-032

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A sound wave has a frequency of 250 Hz and a wavelength of 1.4 m.

Wave speed = frequency $\times$ wavelength

What is the speed of the wave?

- A) 178.6 m/s
- B) 250 m/s
- C) 350 m/s
- D) 1,400 m/s

GED-SCI16-033

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A forest area is cleared for construction, but a protected strip of native shrubs and soil is left between the construction site and a stream. Over time, the strip supports insects, birds, and small mammals while also reducing sediment movement into the stream.

Which ecological role is best described?

- A) Habitat corridor and buffer zone
- B) Food chain with one producer and one consumer
- C) Chromosome separation during meiosis
- D) Combustion of fossil fuels

GED-SCI16-034

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A gene contains instructions for a membrane channel protein that helps ions move across a cell membrane. A mutation changes the channel's shape, so ions move through it less efficiently.

Which statement best explains the effect of the mutation?

- A) A gene change can alter a membrane protein and affect ion movement.
- B) Gene changes cannot affect proteins.
- C) Ion movement occurs only in rocks.
- D) Membrane channels are not made from genetic instructions.

PART B — ANSWER KEY + EXPLANATIONS

GED-SCI16-001 — Correct: A — Correct Answer: Guided breathing was associated with a lower average resting heart rate. — Explanation: Group A used guided breathing and had a lower average resting heart rate than Group B.

GED-SCI16-002 — Correct: B — Correct Answer: Object B — Explanation: Momentum = mass $\times$ velocity. Object B has $5 \text{ kg} \times 4 \text{ m/s} = 20 \text{ kg}\cdot\text{m/s}$, greater than A = 18, C = 16, and D = 18.

GED-SCI16-003 — Correct: A — Correct Answer: Fields with cover crops show higher soil organic matter after several seasons. — Explanation: Cover crops can add organic matter and improve soil quality.

GED-SCI16-004 — Correct: A — Correct Answer: Nervous system — Explanation: The nervous system coordinates rapid responses such as pupil size changes.

GED-SCI16-005 — Correct: 42 — Tolerance: exact — Correct Answer: 42 — Explanation: Momentum = $6 \text{ kg} \times 7 \text{ m/s} = 42 \text{ kg}\cdot\text{m/s}$.

GED-SCI16-006 — Correct: A,B — Correct Answer: Cart speed | Track surface — Explanation: These factors should stay constant so cart mass is the main variable affecting momentum.

GED-SCI16-007 — Correct: D1=recessive inheritance; D2=carrier — Correct Answer: D1=recessive inheritance; D2=carrier — Explanation: A carrier has one changed copy and may pass a recessive allele to children.

GED-SCI16-008 — Correct: A — Correct Answer: Doppler effect — Explanation: The Doppler effect changes observed pitch when a sound source moves relative to an observer.

GED-SCI16-009 — Correct: A — Correct Answer: Seafloor spreading occurs at mid-ocean ridges. — Explanation: Matching magnetic stripes on both sides of a ridge support seafloor spreading.

GED-SCI16-010 — Correct: A — Correct Answer: Average blood oxygen saturation increased after treatment. — Explanation: Average oxygen saturation increased from 91% to 96%.

GED-SCI16-011 — Correct: T1=A; T2=B; T3=C — Correct Answer: T1=Type of wheel tread used; T2=Distance the vehicle travels before stopping; T3=Same starting speed in each trial — Explanation: Wheel tread is changed, stopping distance is measured, and starting speed is controlled.

GED-SCI16-012 — Correct: A — Correct Answer: It helps physically break fat into smaller droplets for digestion. — Explanation: Bile breaks large fat droplets into smaller droplets, increasing the surface area available to enzymes.

GED-SCI16-013 — Correct: A — Correct Answer: Opposite electric charges attract. — Explanation: Materials with opposite electric charges can attract each other.

GED-SCI16-014 — Correct: 25 — Tolerance: exact — Correct Answer: 25 — Explanation: The increase is 300 hectares. $300 \div 1,200 \times 100 = 25\%$.

GED-SCI16-015 — Correct: A — Correct Answer: To compare changes in bone strength over time. — Explanation: Measuring before and after allows researchers to compare how bone strength changes in each group.

GED-SCI16-016 — Correct: A — Correct Answer: Surface material of the slide — Explanation: The independent variable is the factor intentionally changed by the student.

GED-SCI16-017 — Correct: D1=inclined plane; D2=mechanical advantage — Correct Answer: D1=inclined plane; D2=mechanical advantage — Explanation: A ramp is an inclined plane that can reduce the input force needed by increasing distance.

GED-SCI16-018 — Correct: A — Correct Answer: Average systolic blood pressure decreased over the 12 weeks. — Explanation: Average systolic pressure decreased from 146 to 128 mm Hg.

GED-SCI16-019 — Correct: A — Correct Answer: Packed snow — Explanation: The sled traveled farthest on packed snow, suggesting the least friction among the listed surfaces.

GED-SCI16-020 — Correct: A,B — Correct Answer: Average glacier thickness decreased from Year 1 to Year 4. | Average summer temperature increased from Year 1 to Year 4. — Explanation: Glacier thickness decreased from 92 m to 71 m, while summer temperature increased from 10.8°C to 12.3°C.

GED-SCI16-021 — Correct: A — Correct Answer: It can signal tissues involved in growth and repair. — Explanation: Growth hormone can affect tissues involved in growth and repair.

GED-SCI16-022 — Correct: A — Correct Answer: The forces are balanced, so the cabinet moves at constant speed. — Explanation: Equal opposite forces result in no acceleration, so motion continues at constant speed.

GED-SCI16-023 — Correct: A — Correct Answer: Meteor — Explanation: A meteor is the bright streak produced when an object enters Earth's atmosphere.

GED-SCI16-024 — Correct: 16 — Tolerance: exact — Correct Answer: 16 — Explanation: $72 \div 450 \times 100 = 16\%$.

GED-SCI16-025 — Correct: A — Correct Answer: Peak force recorded by the sensor — Explanation: The dependent variable is the measured peak force after changing helmet liner type.

GED-SCI16-026 — Correct: A — Correct Answer: They help cells release usable energy from food molecules. — Explanation: Mitochondria are involved in cellular respiration, which releases usable energy.

GED-SCI16-027 — Correct: A — Correct Answer: The output distance was half the input distance in each trial. — Explanation: In each trial, output distance is one-half of input distance.

GED-SCI16-028 — Correct: D — Correct Answer: Object D — Explanation: Object D has the greatest listed distance from the Sun, 778 million km.

GED-SCI16-029 — Correct: A — Correct Answer: The stretching group had a shorter average recovery time. — Explanation: The stretching group averaged 18 hours compared with 25 hours for the no-stretching group.

GED-SCI16-030 — Correct: A — Correct Answer: Polarization — Explanation: Polarized lenses reduce glare by blocking light vibrating in certain directions.

GED-SCI16-031 — Correct: A,B — Correct Answer: Use the same track for all walking trials. | Measure walking pace the same way each time. — Explanation: Keeping the track and measurement method constant improves the investigation.

GED-SCI16-032 — Correct: C — Correct Answer: 350 m/s — Explanation: Wave speed = $250 \text{ Hz} \times 1.4 \text{ m} = 350 \text{ m/s}$.

GED-SCI16-033 — Correct: A — Correct Answer: Habitat corridor and buffer zone — Explanation: The protected strip provides habitat and helps reduce sediment movement into the stream.

GED-SCI16-034 — Correct: A — Correct Answer: A gene change can alter a membrane protein and affect ion movement. — Explanation: Genes provide instructions for proteins, and mutations can alter protein shape and function.